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🚀 LAB 6 — Resilient Internet + Secure Microservices (Optional)

This lab upgrades Lab 5 from:

> “Working SME network”

to

> “Enterprise-grade resilient WAN + secured microservices”

🎯 LAB 6 Objectives

1. Replace static routing with dynamic routing (OSPF)
2. Add WAN failure detection (IP SLA)
3. Implement basic edge security (ACL firewalling)
4. Harden microservice exposure
5. Add observability foundation
6. Prepare for Phase 2 service scale-out

🗼 LAB 6 Topology Evolution

Same physical layout as Lab 5:

```
Internet
   |
 R1 G0/0 (NAT Edge)
   |
R1 S0/0/0 ---- R2 S0/0/0   (100.10.10.0/30)
   |                      |
LAN A                 LAN B
```

But now we introduce:

- OSPF Area 0 between R1 and R2
- IP SLA tracking default route
- ACL filtering WAN
- Controlled microservice exposure

🔷 PHASE 1 — Replace Static with OSPF

Why?

Static routes:

- No failover intelligence
- No topology awareness
- No scalability

OSPF provides:

- Automatic route exchange
- Convergence
- Metric-based path selection

🔹 R1 OSPF

```
router ospf 1
 router-id 1.1.1.1
 network 100.10.10.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 passive-interface g0/0/1
```

```
Router(config)#router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 100.10.10.0 0.0.0.3 area 0
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# passive-interface g0/0/1
Router(config-router)#
```

◆ R2 OSPF

```
router ospf 1
 router-id 2.2.2.2
 network 100.10.10.0 0.0.0.3 area 0
 network 192.168.20.0 0.0.0.255 area 0
 passive-interface g0/0/1
```

```
Router(config-router)#router ospf 1
Router(config-router)# router-id 2.2.2.2
Router(config-router)# network 100.10.10.0 0.0.0.3 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# passive-interface g0/0/1
Router(config-router)#
```

◆ Default Route Injection from R1

```
router ospf 1
 default-information originate
```

Now:

- R2 automatically learns default route
- No static route needed

```
Router(config-router)#router ospf 1
Router(config-router)# default-information originate
Router(config-router)#
```

◆ PHASE 2 — WAN Failure Detection (IP SLA)

We simulate real ISP outage detection.

On R1

```
ip sla 1
 icmp-echo 8.8.8.8 source-interface g0/0
 frequency 5
```

```
exit

ip sla schedule 1 life forever start-time now

track 1 ip sla 1 reachability

ip route 0.0.0.0 0.0.0.0 g0/0 track 1
```

If Internet fails:

- Default route removed
- OSPF stops advertising it
- R2 loses default cleanly

This mimics enterprise edge behavior.

🔷 PHASE 3 — Basic WAN Security

Currently: WAN is fully open.

We harden it.

🔹 Block unsolicited WAN inbound

On R1:

```
ip access-list extended WAN-IN
 permit tcp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip any any log

interface s0/0/0
 ip access-group WAN-IN in
```

Now:

- Only LAN B → LAN A allowed
- All other WAN inbound blocked
- Logged for visibility

```
Router(config)#ip access-list extended WAN-IN
Router(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config-ext-nacl)# deny ip any any
Router(config-ext-nacl)#exit
Router(config)#interface Serial0/1/0
Router(config-if)# ip access-group WAN-IN in
Router(config-if)#exit
Router(config)#
```

🔷 PHASE 4 — Microservice Exposure Control

Instead of exposing entire server:

Expose only gateway service.

Example:

```
ip nat inside source static tcp 192.168.10.10 8000 interface g0/0 8000
```

Now:

- Only port 8000 reachable from Internet
- Backend services remain private

```
Router(config)#ip nat inside source static tcp 192.168.10.10 8000 203.0.113.1 8000
Router(config)#
```

This models:

- API Gateway pattern
- Real production practice

🔷 PHASE 5 — Observability Foundation

Add:

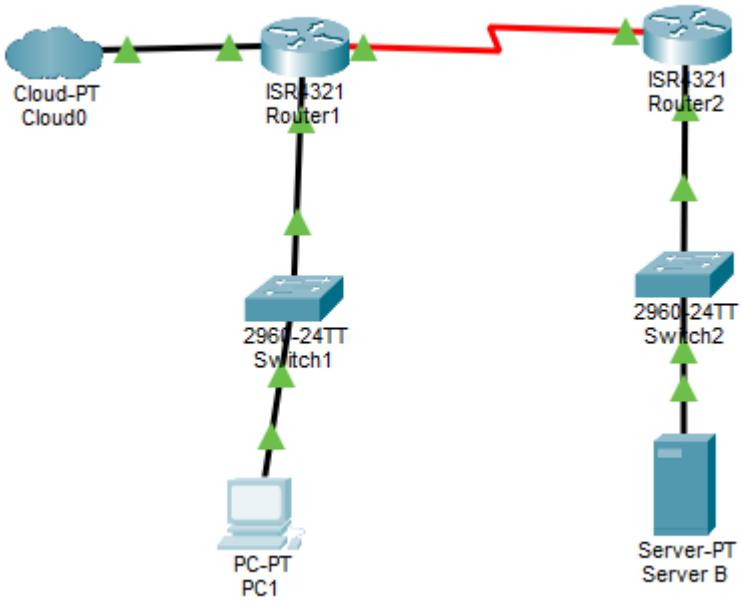
On Routers:

```
service timestamps log datetime msec
logging buffered 4096
```

Enable NetFlow (optional advanced)

```
ip flow-export version 9
```

```
Router(config)#
*Mar 01, 00:38:45.3838: SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml
disabled, filtering disabled, size (4096)
```



🔷 PHASE 6 — Microservice Network Validation

Now validate:

Test 1

ClientA → ServerB via OSPF path

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=6ms TTL=126
Reply from 192.168.20.10: bytes=32 time=5ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>
```

Test 2

R2 loses default if ISP unreachable

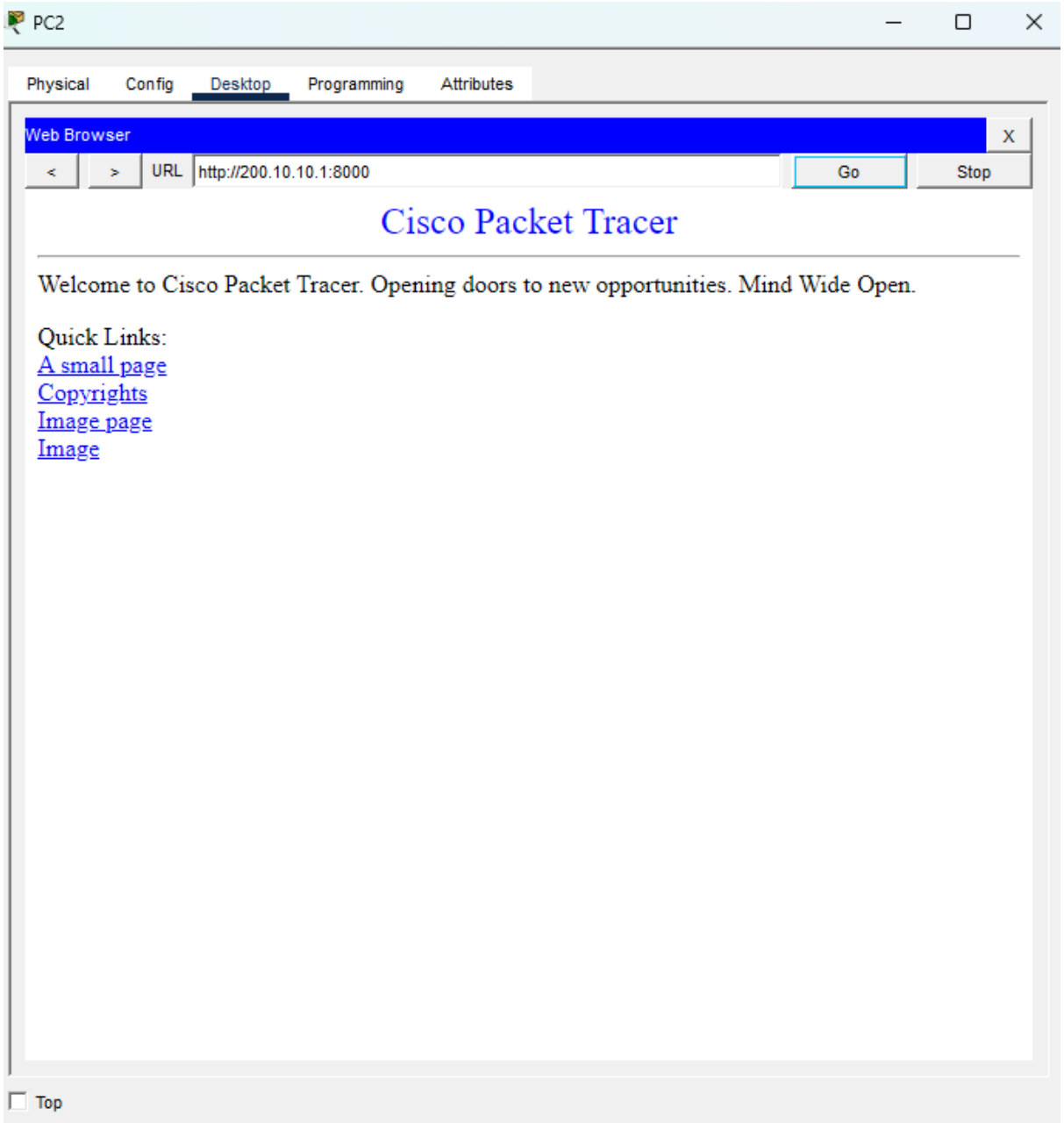
```
Router2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/1/0
L       100.10.10.2/32 is directly connected, Serial0/1/0
O    192.168.10.0/24 [110/65] via 100.10.10.1, 00:51:39, Serial0/1/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/1

Router2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/1/0
L       100.10.10.2/32 is directly connected, Serial0/1/0
O    192.168.10.0/24 [110/65] via 100.10.10.1, 00:52:00, Serial0/1/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/1

Router2>
```

Test 3

Internet cannot access backend ports

Test 4

Gateway reachable from Internet



🧪 LAB 6 Test Plan

Test	Expected Result
OSPF neighbor	FULL
`show ip route ospf`	Remote LAN visible
ISP unplug	Default route disappears
Internet → port 8000	Success

Test	Expected Result
Internet → port 9000	Blocked
LAN A ↔ LAN B	Success

📊 LAB 6 Architecture Classification

Now your topology represents:

✓ Routed WAN with dynamic routing ✓ Conditional default injection ✓ WAN ACL firewall ✓ Controlled service publishing ✓ Production edge logic

This is now:

> Enterprise Branch Architecture v1

📈 Readiness Score After LAB 6

Category	Score
Routing	9/10
WAN Intelligence	9/10
Security	6/10
Microservice Exposure	8/10
Scalability	8/10

Overall:

🟢 8.5 / 10 — Enterprise Branch-Ready

🔮 LAB 7 Preview (If You Want to Continue)

- Dual ISP simulation

- HSRP between R1 and new R3
- Site-to-site VPN
- DMZ zone
- Load balancer in front of microservices
- Redis + PostgreSQL integration
- Central logging stack